

Reasoning: Series, Analogy and Coding-Decoding

तर्कशक्ति: श्रृंखला, सादृश्यता और कूट लेखन-कूट वाचन

How this topic is tested

Topic (iii) of the official syllabus is "Reasoning ability and Quantitative Aptitude". The level is that of the General Ability Test used in UPSC Recruitment Tests (EPFO, CAPF Paper-I and the earlier Vice Principal and Principal papers of 2022), well below CSAT. The verbal-reasoning items are short, mostly one-step questions of the kind covered here: number and letter series, analogy, odd-one-out and coding-decoding. They are the cheapest marks in the paper if you have a fixed method, and the most expensive if you guess, because a wrong answer costs **one-third of the question's marks**.

Three habits for the exam hall:

- Write the alphabet positions on the rough sheet in the first minute (see the table below). Do not do letter arithmetic in your head.
- Check a pattern against **every** given term, not just the first two. UPSC options are built from patterns that fit the first two terms and then break.
- Give a series or coding question **60 to 90 seconds**. If no pattern appears, mark it and move on; come back only if time remains.

Letter-series and coding items are built on the English alphabet (A to Z), so the same tools apply whichever medium you answer in.

The toolkit you must carry

Letter positions

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

- Anchor word **EJOTY**: E = 5, J = 10, O = 15, T = 20, Y = 25. Every other letter is within two steps of an anchor.
- **Opposite letters** add to **27**: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N. Opposite of a letter at position p is at **27 - p**.
- Position from the right end = **27 - position from the left**. The 20th letter from the right is the 7th from the left, G.
- Vowels are A, E, I, O, U (positions 1, 5, 9, 15, 21).

Number facts

Type	Values to recognise on sight
Squares 1 to 20	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400
Cubes 1 to 12	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728
Primes below 50	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 (15 primes; 25 below 100)
$n(n+1)$	2, 6, 12, 20, 30, 42, 56, 72, 90, 110
Factorials	1, 2, 6, 24, 120, 720
Fibonacci	1, 1, 2, 3, 5, 8, 13, 21, 34, 55

Number series

Work in this order and stop as soon as one rule fits all the terms.

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`mermaid flowchart TD
  A[Write the differences] --> B{Constant or in AP?}
  B -- Yes --> C[AP or  $n^2$  rule]
  B -- No --> D[Try ratios]
  D --> E[Compare squares, cubes, primes]
  E --> F[Split into two series]
  C --> G[Check against every term]
  F --> G
  G --> H[ ]
```

Step 1: Differences. Write the gap between consecutive terms.

- Constant gap = arithmetic progression: 7, 12, 17, 22, 27, **32**.
- Gaps themselves in AP = quadratic rule: 2, 7, 14, 23, 34, ? has gaps 5, 7, 9, 11, so the next gap is 13 and the answer is **47**. Equivalently each term is $n^2 + \dots$

(here $n^2 + 2n - 1$); you do not need the formula, the second difference is enough.

- Gaps doubling = multiply-and-add rule: 5, 11, 23, 47, 95 has gaps 6, 12, 24, 48, so the rule is $\times 2 + 1$ and the next term is **191**.

Step 2: Ratios. If the differences do not help, divide.

- Constant ratio = geometric progression: 3, 6, 12, 24, 48, **96**; 5, 15, 45, 135, **405**.
- Increasing multipliers: 1, 2, 6, 24, 120 is $\times 2$, $\times 3$, $\times 4$, $\times 5$, so next is **720** (factorials).
- Mixed: 3, 7, 15, 31, 63 is $\times 2 + 1$, so **127**. 2, 5, 11, 23, 47 is $\times 2 + 1$ as well.

Step 3: Squares, cubes, primes. Compare each term with the nearest square or cube.

Series	Rule	Next
2, 5, 10, 17, 26, 37	$n^2 + 1$	50
0, 3, 8, 15, 24, 35	$n^2 - 1$	48
4, 7, 12, 19, 28, 39	$n^2 + 3$	52
2, 9, 28, 65, 126	$n^3 + 1$	217
100, 81, 64, 49, 36	squares descending	25
6, 10, 14, 22, 26, 34	$2 \times \text{prime}$ (3, 5, 7, 11, 13, 17)	38 (2×19)
1, 1, 2, 3, 5, 8, 13	each term = sum of previous two	21

Step 4: Alternating or two-in-one series. If gaps jump about irregularly (say +17, -14, +11, -8), split the series into odd-position and even-position terms.

- 3, 20, 6, 17, 9, 14, 12, ? splits into 3, 6, 9, 12 and 20, 17, 14, ?, so the answer is **11**.
- 2, 10, 4, 20, 6, 30, 8, ? gives **40**.

Step 5: Wrong-number questions. "Which number does not fit the series?" Find the rule from the majority of terms, then check each. In 2, 5, 10, 17, 26, **38**, 50 the rule is $n^2 + 1$, so 38 should be 37. Typical trap: the wrong term is placed late, so a candidate who checks only the first three terms sees nothing.

Common patterns in one glance

What you see	Try
Slowly growing, gaps constant	AP
Gaps growing by a constant	n^2 type (second difference)
Roughly doubling	$\times 2$ then $\pm$ small number
Roughly tripling	$\times 3$ then $\pm$ small number
Ends in 1, 4, 9, 6, 5, 0 often	squares $\pm k$
2, 3, 5, 7, 11 inside	primes
Up-down-up-down	two interleaved series

Letter series

Convert every letter to its number, then treat it as a number series.

- **Constant gap:** Z, W, T, Q, N, ? is 26, 23, 20, 17, 14, so **11 = K**.
- **Growing gap:** A, C, F, J, O, ? is 1, 3, 6, 10, 15 with gaps 2, 3, 4, 5, so +6 gives **21 = U**. B, D, G, K, P, ? gives **V** in the same way.

- **Opposite-letter alternation:** A, Z, B, Y, C, X, D, ? gives **W**. AZ, BY, CX, DW, ? gives **EV**.
- **Letter groups:** ABC, FGH, KLM, PQR, ? has first letters 1, 6, 11, 16, so the next group starts at 21: **UVW**. AB, DE, GH, JK, ? gives **MN**.
- **Letter-number pairs:** B4, D16, F36, H64, ? is the letter's position squared, so **J100**.
- **Wrap-around:** the alphabet is treated as circular. Counting +3 from Y gives B ($25 + 3 = 28$, $28 - 26 = 2$).

Repeating-pattern series such as _ b c _ c a _ a b: look for a block that repeats. Here the string is **abc / bca / cab**, so the blanks are **a, b, c**. Method: count the total letters including blanks (9), try blocks of 3, then of 2 or 4; test the first full block against the later ones.

Positional questions about the alphabet itself also appear:

- "Which letter is 7th to the right of the 12th letter from the left?" Position $12 + 7 = 19 = \mathbf{S}$.
- "Which letter is 5th to the left of the 10th letter from the right?" 10th from the right is position 17 (Q); 5 to the left is $\mathbf{12 = L}$.
- "If the alphabet is written in reverse order, which letter is 8th from the left?" In reverse order the 8th from the left is the 8th from the right of the normal alphabet: $27 - 8 = 19 = \mathbf{S}$.

Analogy

An analogy question gives a pair with a fixed relationship and asks you to complete a second pair

with **the same relationship in the same direction**. Say the relation in words first, then apply it.

Verbal analogy: relationship types

Relationship	Example	Completion
Worker : tool	Sculptor : Chisel :: Writer : ?	Pen
Worker : workplace	Doctor : Hospital :: Teacher : ?	School
Study : subject	Botany : Plants :: Entomology : ?	Insects
Instrument : quantity	Barometer : Pressure :: Anemometer : ?	Wind speed
Part : whole	Page : Book :: Petal : ?	Flower
Object : function	Pen : Write :: Knife : ?	Cut
Group name	Flock : Sheep :: Herd : ?	Cattle
Product : raw material	Paper : Pulp :: Cloth : ?	Yarn
Degree (intensity)	Warm : Hot :: Cool : ?	Cold
Cause : effect	Germ : Disease :: Fire : ?	Smoke
Male : female	Horse : Mare :: Dog : ?	Bitch
Young one	Cow : Calf :: Dog : ?	Puppy
Country : currency	Japan : Yen :: Bangladesh : ?	Taka
State : capital	Kerala : Thiruvananthapuram :: Odisha : ?	Bhubaneswar

Study-subject pairs UPSC likes: **Ornithology** (birds), **Entomology** (insects), **Ichthyology** (fishes), **Herpetology** (reptiles), **Pedology** (soils), **Seismology** (earthquakes), **Oology** (eggs), **Mycology** (fungi), **Pedagogy** (teaching). Instruments: **Barometer** (atmospheric pressure), **Hygrometer** (humidity), **Anemometer** (wind speed), **Seismograph** (earthquakes), **Sphygmomanometer** (blood pressure), **Lactometer** (purity of milk), **Odometer** (distance travelled).

Hindi-medium versions use the same logic: कवि : कविता :: उपन्यासकार : उपन्यास; चिकित्सक : अस्पताल :: शिक्षक : विद्यालय.

Number analogy

Find an operation that converts the first number into the second, then apply it to the third.

Pair	Rule	Apply
7 : 50 :: 11 : ?	$n^2 + 1$	122
8 : 72 :: 12 : ?	$n(n + 1)$	156
4 : 64 :: 5 : ?	n^3	125
3 : 11 :: 7 : ?	$n^2 + 2$	51
6 : 42 :: 9 : ?	$n(n + 1)$	90
12 : 144 :: 15 : ?	n^2	225
5 : 36 :: 7 : ?	$(n + 1)^2$	64

Trap: several rules may fit a single pair (7 : 50 fits $n^2 + 1$, i.e. $7 \times 7 + 1$, and also $n(n + 1) - 6$, i.e. $7 \times 8 - 6$, which would give 126 for 11). Choose the simplest rule, and check whether the options include the result of a

competing rule (here 121 from n^2 and 132 from $n(n + 1)$) would be distractors). When two pairs are given, the rule must fit both.

Letter analogy

- **Same shift:** CAT : DBU :: DOG : ? is +1 on each letter, so **EPH**. AEG : CGI :: KOQ : ? is +2 on each letter, so **MQS**.
- **Opposite letters:** ACE : ZXV :: BDF : ? replaces each letter by its opposite, so **YWU**.
- **Reverse order:** READ : DAER :: BOOK : ? gives **KOOB**.
- **Different shift per position:** AEG : BGJ :: KOQ : ? is +1, +2, +3, so **LQT**.

Classification or odd one out

Three items share a property; one does not. Check in this order: category, then number property (prime, square, even/odd, digit sum), then letter property (vowels, position gaps).

- 17, 23, 29, **39**: all primes except 39 (3×13).
- 121, 144, 169, **190**: all perfect squares except 190.
- Copper, Zinc, **Brass**, Tin: Brass is an alloy, the rest are elements.
- Ganga, Yamuna, **Sambhar**, Kaveri: Sambhar is a lake, the rest are rivers.
- BDF, HJL, **NPS**, TVX: every other group has gaps of +2, +2; NPS has +2, +3.

Coding-decoding

A word is turned into a code by a rule; you must find the rule and apply it (coding) or reverse it (decoding).

Write the positions under both the word and its code; the rule shows up as a column of differences.

Type 1: Constant letter shift

- SCHOOL = TDIPPM: every letter +1. So TEACHER = **UFBDIFS**.
- DELHI = EFMIJ (+1), so PATNA = **QBUOB**.
- Backward shift: TEACHER with -1 = **SDZBGDQ**.
- Wrap-around applies: with +2, Y becomes A and Z becomes B.

Type 2: Variable shift

The shift changes with position: +1, +2, +3, ... or +1, -1, +1, -1.

- CAT = DCW (+1, +2, +3), so DOG = **EQJ** (D+1 = E, O+2 = Q, G+3 = J).
- Always verify the pattern on all three letters of the sample word before applying it.

Type 3: Opposite-letter (reversal) coding

Each letter is replaced by its opposite (positions add to 27).

- RAM = IZN (R 18 to I 9, A to Z, M 13 to N 14). So HOPE = **SLKV** and CAT = **XZG**.
- A quick check for this type: if A becomes Z or M becomes N anywhere in the sample, it is the opposite-letter code.

Type 4: Rearrangement

- Whole word reversed: SCHOOL = LOOHCS.

- Pairs swapped: MOTHER = **OMHTRE** (MO to OM, TH to HT, ER to RE). A word with an odd number of letters keeps its last letter in place.
- Reversed and then shifted: combine the two steps in the order the sample shows.

Type 5: Number coding

- Position values written out: BAD = **214**, CAB = 312.
- Sum of positions: CAT = 3 + 1 + 20 = **24**, DOG = 4 + 15 + 7 = 26.
- Position values with a constant added or multiplied: if A = 2, B = 4, C = 6 (double the position), then DOG = 8, 30, 14.
- Codes using digits for letters can also run backwards (A = 26, Z = 1).

Type 6: Substitution (word-for-word) coding

"If 'pen' is called 'pencil', 'pencil' is called 'book', 'book' is called 'bag', and 'bag' is called 'chalk', what does a student carry books in?" The real answer is a **bag**, and a bag is now called **chalk**. Rule: first answer the question in plain language, then translate that one word into its new name. Never translate the words in the question itself.

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`mermaid flowchart TD
  A[Read the question as English] --> B[Real answer: bag]
  B --> C[Find new name of bag]
  C --> D[Bag is called chalk]
  D --> E[Mark: chalk]`
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Colour version: "If sky is called green, green is called red and red is called yellow, what is the colour of grass?" Grass is green; green is called **red**.

Type 7: Sentence (language) coding

Each word has a code word, but the order inside a sentence may change. Find words common to two sentences; the common code word belongs to the common English word.

- 'ne pa ki' = students are bright; 'ki la mo' = bright sun shines; 'ne ro ta' = students study hard.
- 'ki' is common to sentences 1 and 2, so ki = bright. 'ne' is common to 1 and 3, so ne = students. The remaining word in sentence 1, **pa = are**.
- If a word appears in only one sentence, its code can only be found by elimination, and sometimes the honest answer is "cannot be determined".

Type 8: Mathematical operator coding

The symbols are swapped; rewrite the expression with the real operators and then apply **BODMAS** (division and multiplication before addition and subtraction, left to right).

- If + means $\times$, - means $\div$, $\times$ means +, $\div$ means -, then $8 + 4 - 2 \times 6 \div 3$ becomes $8 \times 4 \div 2 + 6 - 3 = 16 + 6 - 3 = 19$.
- Also seen: "If $2 \times 3 = 49$ and $4 \times 5 = 1625$, then $5 \times 6 = ?$ " (write the squares side by side: $2^2 = 4$ and $3^2 = 9$ give 49, so 5×6 gives **2536**).

Decoding checklist

1. Write positions under word and code.
2. Look at the differences: all the same (Type 1), changing (Type 2), summing to 27 (Type 3).

3. If letters are the same but in a new order, it is rearrangement (Type 4).
4. If the code is numbers, test position values and their sum (Type 5).
5. For sentences, use common words (Type 7).

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``mermaid flowchart TD
  A[Write positions under both] --> B{What does the code look like?}
  B -- Letters, differences same --> C[Type 1 constant shift]
  B -- Letters, differences changing --> D[Type 2 variable shift]
  B -- Positions sum to 27 --> E[Type 3 opposite letters]
  B -- Same letters, new order --> F[Type 4 rearrangement]
  B -- Numbers --> G[Type 5 number coding]
  B -- Whole sentences --> H[Type 7 use common words]
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Speed plan for the paper

- Reasoning items first, then arithmetic, then data tables. These three types should take under 90 seconds each.
- For series, write the differences in a row below the terms; for letter work, write positions. Rough work prevents the one-letter slip that UPSC's distractors are built on (options such as UFBDJFS beside the correct UFBDIFS).
- When exactly one option fits your rule and you have checked the rule against every term, answer. If two rules fit and give different options, choose the simpler rule; if you cannot decide, skip. A blind guess with the one-third penalty has **zero expected value**.

Common traps

1. **Pattern checked on two terms only.** 2, 6, 12, 20, 30 followed by 40 is wrong; the gaps grow (4, 6, 8, 10), so the answer is 42.
2. **Constant-gap assumption in letter series.** B, D, G, K, P continues with V (gap +6), not U.
3. **Direction of the analogy.** Thermometer : Temperature means instrument : quantity. The answer to Hygrometer : ? is Humidity, not "Weather".
4. **Competing number-analogy rules.** 7 : 50 could be $n^2 + 1$ (gives 122 for 11) or other rules; the options will include 121 and 132 as traps.
5. **One letter off in coding options.** UFBDIFS, UFBDJFS and UFBCIFS look alike; write each letter's position.
6. **Wrap-around ignored.** Y + 3 is B, not an error.
7. **Substitution coding answered with the real word.** The answer is the new name of the real thing (chalk, not bag).
8. **Positions from the right.** The 20th from the right is $27 - 20 = 7$ th from the left, G. Forgetting the +1 gives F.
9. **Opposite letter miscounted.** Use $27 - p$; M and N are each other's opposite.
10. **Operator coding without BODMAS.** After substituting, do division and multiplication first.
11. **Wrong-number series.** The misfit is usually near the end; test every term.
12. **Odd one out by appearance.** 39 looks prime but equals 3×13 ; 91 equals 7×13 .

13. **Sentence coding word order.** Code words are not in the same order as the English words; use only common words.

Remember for the exam

1. Letter positions: EJOTY = 5, 10, 15, 20, 25; write the full table on rough paper first.
2. Opposite letters add to 27; the opposite of position p is $27 - p$ (A-Z, M-N).
3. Position from the right = $27 -$ position from the left.
4. Series order of attack: differences, ratios, squares and cubes, primes, split into two series.
5. Gaps in AP mean an n^2 pattern; gaps doubling mean $\times 2$ plus or minus a constant.
6. $n(n+1)$ series: 2, 6, 12, 20, 30, 42, 56. $n^2 + 1$: 2, 5, 10, 17, 26, 37, 50.
7. Primes below 50: 15 of them; below 100: 25; 2 is the only even prime; 1 is not prime.
8. Analogy: name the relationship in words, keep the same direction, then apply.
9. Instruments: barometer pressure, hygrometer humidity, anemometer wind speed, lactometer milk.
10. Coding: $+k$ shift, variable shift, opposite letters (sum 27), rearrangement, number codes, substitution, sentence codes, operator swaps.
11. Substitution coding: answer with the new name of the real object.
12. Sentence coding: the code for a word is the code common to all sentences containing that word.

13. Operator coding: substitute, then BODMAS.
14. Alphabet is circular: shifts wrap from Z back to A.
15. Check every term before marking; a wrong answer costs one-third of the marks, a blank costs nothing.

Quick facts

- Letter positions anchor EJOTY: E = 5, J = 10, O = 15, T = 20, Y = 25; write the full A-Z table on rough paper first.
- Opposite letters add to 27 (A-Z, B-Y, M-N); opposite of position $p = 27 - p$.
- Position from the right end of the alphabet = $27 - \text{position from the left}$; 20th from the right is G.
- Number series attack order: differences, then ratios, then squares/cubes/primes, then split into two interleaved series.
- Gaps forming an AP mean an n-squared pattern; gaps doubling mean multiply by 2 then add or subtract a constant.
- $n(n+1)$: 2, 6, 12, 20, 30, 42, 56; n squared + 1: 2, 5, 10, 17, 26, 37, 50; n squared - 1: 0, 3, 8, 15, 24, 35, 48.
- Primes below 50 are 15 in number, below 100 are 25; 2 is the only even prime and 1 is not prime.
- Analogy method: state the relationship in words (worker : tool, study : subject, instrument : quantity), keep the same direction, then apply it to the third term.

- Instruments: barometer measures pressure, hygrometer humidity, anemometer wind speed, lactometer purity of milk, sphygmomanometer blood pressure.
- Letter-shift coding: SCHOOL = TDIPPM is +1, so TEACHER = UFBDIFS; the alphabet wraps from Z to A.
- Opposite-letter coding: RAM = IZN, so HOPE = SLKV and CAT = XZG.
- Substitution coding: answer with the new name of the real object (a bag called 'chalk' is the answer 'chalk').
- Sentence coding: the code of a word is the code word common to every sentence containing that word.
- Operator coding: replace the symbols with the real operators, then apply BODMAS.
- A wrong answer costs one-third of the question's marks; check the rule on every term before marking, and leave the question blank rather than guess blindly.